



VII Semester (CSE/ IT) REMIDTERM TEST
DEEP LEARNING (CSE_4064/IT_4442)

Time Duration: 2 Hours

Date:

Max marks: 30 MARKS

Question No	Topic	Marks	BL	CO
1.	Which of the following is the most suitable activation function for solving the vanishing gradient problem in deep networks? a) Sigmoid b) Tanh ☒ c) ReLU d) Softmax	1	1	1
2.	Which of the following is true about the decision tree algorithm? a) It always provides the most accurate predictions ☒ b) It is prone to overfitting on large datasets c) It is a type of unsupervised learning algorithm d) It can only handle binary classification problems	1	1	1
3.	Which of the following best describes the purpose of batch normalization in deep learning? a) To prevent overfitting by reducing the number of features ☒ b) To ensure that all features are scaled to a similar range c) To improve the model's ability to generalize by randomly dropping units d) To reduce the training time by increasing the batch size	1	3	2
4.	Which optimization technique is most effective for training deep networks with a high risk of overfitting? a) Adam b) Stochastic Gradient Descent (SGD) c) RMSprop ☒ d) L2 Regularization	1	3	1
5.	Which method would most effectively reduce the risk of overfitting in deep learning models? a) Increasing model depth b) Reducing training data ☒ c) Applying dropout regularization d) Removing the activation function	1	2	1
	Which of the following methods can be used to prevent overfitting in a decision tree model? ☒ a) Decreasing the depth of the tree ☒ b) Increasing the number of features used for splitting c) Using a larger training dataset	1	2	2

6.	d) Reducing the number of training iterations				
7.	Which of the following techniques would best help to prevent vanishing gradients during backpropagation in deep neural networks? a) Using a high learning rate ☒ b) Implementing batch normalization c) Adding more hidden layers d) Reducing the number of neurons per layer	1	3	2	1
8.	In the context of deep learning, what is the primary benefit of using a dropout technique? ☒ a) To prevent overfitting by randomly disabling units during training b) To reduce computation by reducing model size c) To increase the capacity of the model by adding extra hidden layers d) To speed up training by increasing the learning rate	1	2	3	1
9.	In the context of supervised learning, which of the following would typically be used as a loss function for a regression problem? a) Cross-entropy loss b) Hinge loss ☒ c) Mean squared error (MSE) d) Kullback-Leibler divergence	1	2	2	1
10.	Which of the following techniques is primarily used to address the issue of class imbalance in a classification problem? a) Feature scaling b) Data augmentation ☒ c) SMOTE (Synthetic Minority Over-sampling Technique) d) Dimensionality reduction	1	2	1	1
11(a)	You are given a dataset of housing prices, which includes features such as the size of the house, the number of bedrooms, the neighborhood rating, and the year it was built. Your task is to build a machine learning model to predict housing prices. Construct a model evaluation strategy that uses cross-validation to assess the performance of your housing price prediction model.	5	3	3	
11(b)	Analyze the role of activation functions in a feedforward neural network. What would be the effect on model learning if no activation functions were used between the layers?	5	4	1	
12(a)	Apply the backpropagation algorithm to a neural network. How would you calculate the gradient and update the weights for each layer during training?	5	3	2	
12(b)	Compare the performance of k-NN with other classification algorithms, such as decision trees or support vector machines. What factors should be considered when deciding whether k-NN is the best option?	5	4	3	

Note: BL refers to Bloom's Taxonomy Level.